

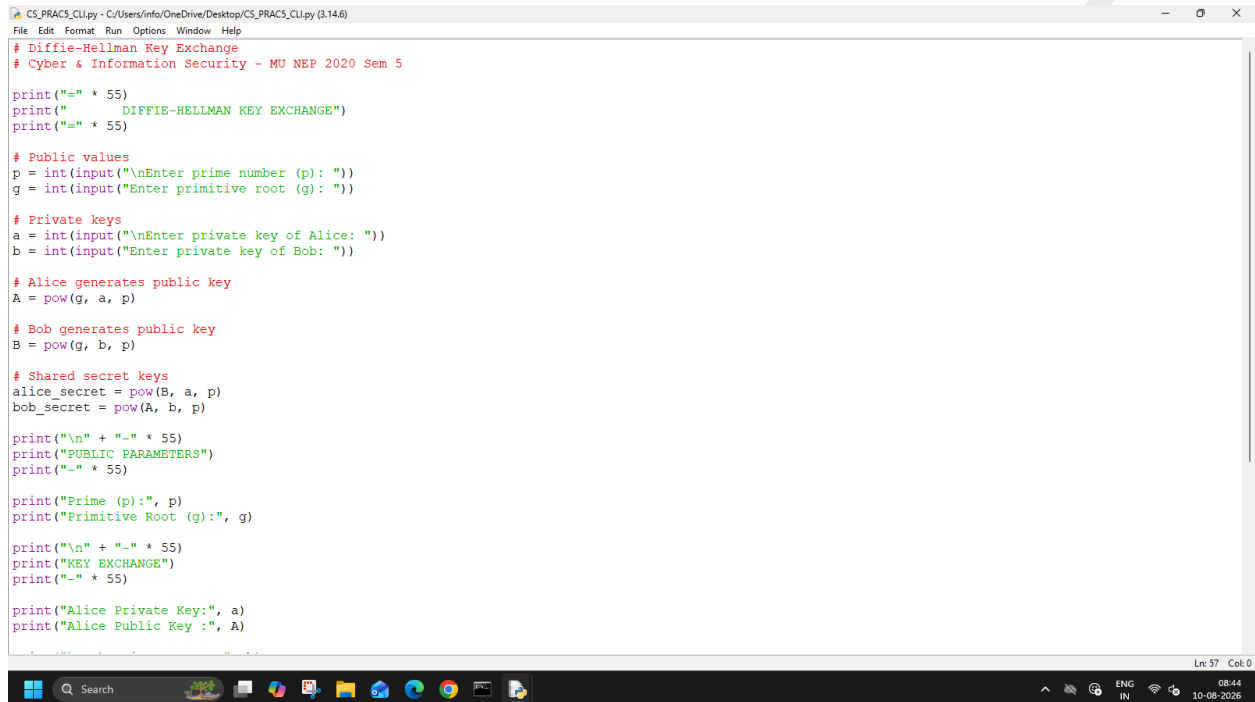
SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT: CYBER & INFORMATION SECURITY

PRACTICAL - 5

AIM: Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

CLI:

Code:



```
CS_PRAC5_CLI.py - C:/Users/info/OneDrive/Desktop/CS_PRAC5_CLI.py (3.14.6)
File Edit Format Run Options Window Help
# Diffie-Hellman Key Exchange
# Cyber & Information Security - MU NEP 2020 Sem 5

print("=" * 55)
print("      DIFFIE-HELLMAN KEY EXCHANGE")
print("=" * 55)

# Public values
p = int(input("\nEnter prime number (p): "))
g = int(input("Enter primitive root (g): "))

# Private keys
a = int(input("\nEnter private key of Alice: "))
b = int(input("Enter private key of Bob: "))

# Alice generates public key
A = pow(g, a, p)

# Bob generates public key
B = pow(g, b, p)

# Shared secret keys
alice_secret = pow(B, a, p)
bob_secret = pow(A, b, p)

print("\n" + "-" * 55)
print("PUBLIC PARAMETERS")
print("-" * 55)

print("Prime (p):", p)
print("Primitive Root (g):", g)

print("\n" + "-" * 55)
print("KEY EXCHANGE")
print("-" * 55)

print("Alice Private Key:", a)
print("Alice Public Key :", A)
```

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```
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# Bob generates public key
B = pow(g, b, p)

# Shared secret keys
alice_secret = pow(B, a, p)
bob_secret = pow(A, b, p)

print("\n" + "-" * 55)
print("PUBLIC PARAMETERS")
print("-" * 55)

print("Prime (p):", p)
print("Primitive Root (g):", g)

print("\n" + "-" * 55)
print("KEY EXCHANGE")
print("-" * 55)

print("Alice Private Key:", a)
print("Alice Public Key :", A)

print("\nBob Private Key :", b)
print("Bob Public Key  :", B)

print("\n" + "-" * 55)
print("SHARED SECRET")
print("-" * 55)

print("Alice's Shared Secret:", alice_secret)
print("Bob's Shared Secret  :", bob_secret)

if alice_secret == bob_secret:
    print("\n✓ KEY EXCHANGE SUCCESSFUL")
    print("Both Alice and Bob have the same secret key.")
else:
    print("\n✗ KEY EXCHANGE FAILED")

print("-" * 55)
```

OUTPUT:

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help

Python 3.14.6 (tags/v3.14.6:c63a6c6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/info/OneDrive/Desktop/CS_PRAC5_CLI.py =====

DIFFIE-HELLMAN KEY EXCHANGE
=====

Enter prime number (p): 3
Enter primitive root (g): 16

Enter private key of Alice: 420
Enter private key of Bob: 124

=====
PUBLIC PARAMETERS
=====
Prime (p): 3
Primitive Root (g): 16

=====
KEY EXCHANGE
=====
Alice Private Key: 420
Alice Public Key : 1

Bob Private Key : 124
Bob Public Key  : 1

=====
SHARED SECRET
=====
Alice's Shared Secret: 1
Bob's Shared Secret  : 1

✓ KEY EXCHANGE SUCCESSFUL
Both Alice and Bob have the same secret key.
=====

>>>
```

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```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Info>python "C:\Users\Info\OneDrive\Desktop\CS_PRACS_CLI.py"
=====
DIFFIE-HELLMAN KEY EXCHANGE
=====

Enter prime number (p): 3
Enter primitive root (g): 13
Enter private key of Alice: 5
Enter private key of Bob: 6

=====
PUBLIC PARAMETERS
=====
Prime (p): 3
Primitive Root (g): 13

=====
KEY EXCHANGE
=====
Alice Private Key: 5
Alice Public Key : 1

Bob Private Key : 6
Bob Public Key  : 1

=====
SHARED SECRET
=====
Alice's Shared Secret: 1
Bob's Shared Secret  : 1

// KEY EXCHANGE SUCCESSFUL
Both Alice and Bob have the same secret key.
=====

C:\Users\Info>
```

SUMEET YADAV
T125

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GUI:

CODE:

```
CS_PRACS_GUI.py - C:/Users/info/OneDrive/Desktop/CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help
import tkinter as tk
from tkinter import messagebox

# DIFFIE-HELLMAN KEY EXCHANGE
def calculate_exchange():
    try:
        p = int(p_entry.get())
        g = int(g_entry.get())
        a = int(alice_private_entry.get())
        b = int(bob_private_entry.get())

    except ValueError:
        messagebox.showerror(
            "Invalid Input",
            "Please enter valid integer values."
        )
        return

    if p <= 1:
        messagebox.showerror(
            "Invalid Prime",
            "Prime number must be greater than 1."
        )
        return

    if a <= 0 or b <= 0:
        messagebox.showerror(
            "Invalid Private Key",
            "Private keys must be positive."
        )
        return

    # Alice's public key
    alice_public = pow(g, a, p)

    # Bob's public key
    bob_public = pow(g, b, p)

    # Shared secrets
    alice_secret = pow(
        bob_public,
        a,
        p
    )

Ln 11 Col 0
```

```
CS_PRACS_GUI.py - C:/Users/info/OneDrive/Desktop/CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

        bob_public,
        a,
        p
    )

    bob_secret = pow(
        alice_public,
        b,
        p
    )

    # Display Alice
    alice_public_value.config(
        text=str(alice_public)
    )

    alice_secret_value.config(
        text=str(alice_secret)
    )

    # Display Bob
    bob_public_value.config(
        text=str(bob_public)
    )

    bob_secret_value.config(
        text=str(bob_secret)
    )

    # Verify
    if alice_secret == bob_secret:
        result_title.config(
            text="✓ KEY EXCHANGE SUCCESSFUL",
            fg="#22c55e"
        )

        result_message.config(
            text="Alice and Bob generated the same shared secret key."
        )

        status_label.config(
            text="● Secure shared key established",
            fg="#22c55e"
        )

Ln 11 Col 0
```

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```
CS_PRACS_GUI.py - C:/Users/info/OneDrive/Desktop/CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

else:

    result_title.config(
        text="X KEY EXCHANGE FAILED",
        fg="#ef4444"
    )

    result_message.config(
        text="The shared secret keys do not match."
    )

    status_label.config(
        text="● Key exchange failed",
        fg="#ef4444"
    )

def clear_all():

    p_entry.delete(0, tk.END)
    g_entry.delete(0, tk.END)

    alice_private_entry.delete(0, tk.END)
    bob_private_entry.delete(0, tk.END)

    alice_public_value.config(text="--")
    alice_secret_value.config(text="--")

    bob_public_value.config(text="--")
    bob_secret_value.config(text="--")

    result_title.config(
        text="READY",
        fg="#60a5fa"
    )

    result_message.config(
        text="Enter the parameters and start the key exchange."
    )

    status_label.config(
        text="● Ready",
        fg="#94a3b8"
    )

Ln: 11 Col: 0
```

```
CS_PRACS_GUI.py - C:/Users/info/OneDrive/Desktop/CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

    status_label.config(
        text="● Ready",
        fg="#94a3b8"
    )

# GUI
root = tk.Tk()

root.title(
    "Diffie-Hellman Key Exchange"
)

root.geometry(
    "950x720"
)

root.resizable(
    False,
    False
)

root.configure(
    bg="#0b1120"
)

# HEADER
header = tk.Frame(
    root,
    bg="#111827",
    height=100
)

header.pack(
    fill="x"
)

header.pack_propagate(False)

tk.Label(
    header,
    text="🔒 DIFFIE-HELLMAN KEY EXCHANGE",
    font=("Arial", 24, "bold")
)

Ln: 11 Col: 0
```

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```
CS_PRACS_GUI.py - C:/Users/info/OneDrive/Desktop/CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

    fg="white"
).pack(
    pady=(18, 5)
)

tk.Label(
    header,
    text="Cyber & Information Security | MU NEP 2020 | Semester 5",
    font=("Arial", 10),
    bg="#111827",
    fg="#94a3b8"
).pack()

# PUBLIC PARAMETERS
parameter_frame = tk.LabelFrame(
    root,
    text=" PUBLIC PARAMETERS ",
    font=("Arial", 11, "bold"),
    bg="#111827",
    fg="white",
    padx=15,
    pady=10
)

parameter_frame.pack(
    fill="x",
    padx=30,
    pady=15
)

# Prime
tk.Label(
    parameter_frame,
    text="Prime Number (p)",
    font=("Arial", 10, "bold"),
    bg="#111827",
    fg="#cbd5e1"
).grid(
    row=0,
    column=0,
    padx=10,
    pady=5
)

Ln: 11 Col: 0
08:53
10-08-2026
```

```
CS_PRACS_GUI.py - C:/Users/info/OneDrive/Desktop/CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

p_entry = tk.Entry(
    parameter_frame,
    font=("Consolas", 11),
    width=18,
    bg="#0f172a",
    fg="white",
    insertbackground="white",
    relief="flat"
)

p_entry.grid(
    row=1,
    column=0,
    padx=10,
    pady=5
)

p_entry.insert(
    0,
    "23"
)

# Primitive root
tk.Label(
    parameter_frame,
    text="Primitive Root (g)",
    font=("Arial", 10, "bold"),
    bg="#111827",
    fg="#cbd5e1"
).grid(
    row=0,
    column=1,
    padx=10,
    pady=5
)

g_entry = tk.Entry(
    parameter_frame,
    font=("Consolas", 11),
    width=18
)

Ln: 11 Col: 0
08:54
10-08-2026
```

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```
CS_PRACS_GUI.py - C:/Users/info/OneDrive/Desktop/CS_PRACS_GUI.py (3.14.6)
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fg="white",
insertbackground="white",
relief="flat"
)

g_entry.grid(
    row=1,
    column=1,
    padx=10,
    pady=5
)

g_entry.insert(
    0,
    "5"
)

# ALICE AND BOB
people_frame = tk.Frame(
    root,
    bg="#0b1120"
)

people_frame.pack(
    fill="x",
    padx=30
)

# ALICE
alice_frame = tk.LabelFrame(
    people_frame,
    text="👤 ALICE ",
    font=("Arial", 12, "bold"),
    bg="#111827",
    fg="#60a5fa",
    padx=20,
    pady=10
)

alice_frame.pack(
    side="left",
    fill="both",
    expand=True,
    padx=(0, 10)
)

Ln 11 Col 0
08:54
10-08-2026
```

```
CS_PRACS_GUI.py - C:/Users/info/OneDrive/Desktop/CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

side="left",
fill="both",
expand=True,
padx=(0, 10)
)

tk.Label(
    alice_frame,
    text="Private Key (a)",
    font=("Arial", 10),
    bg="#111827",
    fg="#cbd5e1"
).pack(
    anchor="w"
)

alice_private_entry = tk.Entry(
    alice_frame,
    font=("Consolas", 11),
    bg="#0f172a",
    fg="white",
    insertbackground="white",
    relief="flat"
)

alice_private_entry.pack(
    fill="x",
    pady=5
)

alice_private_entry.insert(
    0,
    "6"
)

tk.Label(
    alice_frame,
    text="Public Key (A)",
    font=("Arial", 10),
    bg="#111827",
    fg="#cbd5e1"
)

Ln 11 Col 0
08:55
10-08-2026
```

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```
CS_PRACS_GUI.py - C:/Users/info/OneDrive/Desktop/CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

    anchor="w",
    pady=(8, 0)
)

alice_public_value = tk.Label(
    alice_frame,
    text="-",
    font=("Consolas", 16, "bold"),
    bg="#0f172a",
    fg="#60a5fa",
    anchor="w",
    padx=10,
    pady=8
)

alice_public_value.pack(
    fill="x",
    pady=5
)

tk.Label(
    alice_frame,
    text="Shared Secret",
    font=("Arial", 10),
    bg="#111827",
    fg="#cbd5e1"
).pack(
    anchor="w",
    pady=(8, 0)
)

alice_secret_value = tk.Label(
    alice_frame,
    text="-",
    font=("Consolas", 16, "bold"),
    bg="#0f172a",
    fg="#22c55e",
    anchor="w",
    padx=10,
    pady=8
)

Ln: 11 Col: 0
```

```
CS_PRACS_GUI.py - C:/Users/info/OneDrive/Desktop/CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

    alice_secret_value.pack(
        fill="x",
        pady=5
    )
)

#BOB
bob_frame = tk.LabelFrame(
    people_frame,
    text=" @ BOB ",
    font=("Arial", 12, "bold"),
    bg="#111827",
    fg="#c084fc",
    padx=20,
    pady=10
)

bob_frame.pack(
    side="left",
    fill="both",
    expand=True,
    padx=(10, 0)
)

tk.Label(
    bob_frame,
    text="Private Key (b)",
    font=("Arial", 10),
    bg="#111827",
    fg="#cbd5e1"
).pack(
    anchor="w"
)

bob_private_entry = tk.Entry(
    bob_frame,
    font=("Consolas", 11),
    bg="#0f172a",
    fg="white",
    insertbackground="white",
    relief="flat"
)

Ln: 11 Col: 0
```


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```
CS_PRACS_GUI.py - C:\Users\info\OneDrive\Desktop\CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

bob_private_entry.pack(
    fill="x",
    pady=5
)

bob_private_entry.insert(
    0,
    "15"
)

tk.Label(
    bob_frame,
    text="Public Key (B)",
    font=("Arial", 10),
    bg="#111827",
    fg="#cbd5e1"
).pack(
    anchor="w",
    pady=(8, 0)
)

bob_public_value = tk.Label(
    bob_frame,
    text="-",
    font=("Consolas", 16, "bold"),
    bg="#0f172a",
    fg="#c084fc",
    anchor="w",
    padx=10,
    pady=8
)

bob_public_value.pack(
    fill="x",
    pady=5
)

tk.Label(
    bob_frame,
    text="Shared Secret",
    font=("Arial", 10)

```

```
CS_PRACS_GUI.py - C:\Users\info\OneDrive\Desktop\CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

people_frame,
text=" ALICE ",
font=("Arial", 12, "bold"),
bg="#111827",
fg="#60a5fa",
padx=20,
pady=10
)

alice_frame.pack(
    side="left",
    fill="both",
    expand=True,
    padx=(0, 10)
)

tk.Label(
    alice_frame,
    text="Private Key (a)",
    font=("Arial", 10),
    bg="#111827",
    fg="#cbd5e1"
).pack(
    anchor="w"
)

alice_private_entry = tk.Entry(
    alice_frame,
    font=("Consolas", 11),
    bg="#0f172a",
    fg="white",
    insertbackground="white",
    relief="flat"
)

alice_private_entry.pack(
    fill="x",
    pady=5
)

alice_private_entry.insert(
    0

```

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```
CS_PRACS_GUI.py - C:\Users\info\OneDrive\Desktop\CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

font=("Consolas", 11),
bg="#0f172a",
fg="white",
insertbackground="white",
relief="flat"
)

alice_private_entry.pack(
    fill="x",
    pady=5
)

alice_private_entry.insert(
    0,
    "6"
)

tk.Label(
    alice_frame,
    text="Public Key (A)",
    font=("Arial", 10),
    bg="#111827",
    fg="#cbd5e1"
).pack(
    anchor="w",
    pady=(8, 0)
)

alice_public_value = tk.Label(
    alice_frame,
    text="-",
    font=("Consolas", 16, "bold"),
    bg="#0f172a",
    fg="#60a5fa",
    anchor="w",
    padx=10,
    pady=8
)

alice_public_value.pack(
    fill="x",
    pady=5
)

Ln: 1 Col: 0
```

```
CS_PRACS_GUI.py - C:\Users\info\OneDrive\Desktop\CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

alice_public_value.pack(
    fill="x",
    pady=5
)

tk.Label(
    alice_frame,
    text="Shared Secret",
    font=("Arial", 10),
    bg="#111827",
    fg="#cbd5e1"
).pack(
    anchor="w",
    pady=(8, 0)
)

alice_secret_value = tk.Label(
    alice_frame,
    text="-",
    font=("Consolas", 16, "bold"),
    bg="#0f172a",
    fg="#22c55e",
    anchor="w",
    padx=10,
    pady=8
)

alice_secret_value.pack(
    fill="x",
    pady=5
)

#BOB
bob_frame = tk.LabelFrame(
    people_frame,
    text=" @ BOB ",
    font=("Arial", 12, "bold"),
    bg="#111827",
    fg="#c084fc",
    padx=20
)

Ln: 1 Col: 0
```

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```
CS_PRACS_GUI.py - C:\Users\info\OneDrive\Desktop\CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

    fg="white",
    insertbackground="white",
    relief="flat"
)

bob_private_entry.pack(
    fill="x",
    pady=5
)

bob_private_entry.insert(
    0,
    "15"
)

tk.Label(
    bob_frame,
    text="Public Key (B)",
    font=("Arial", 10),
    bg="#11827",
    fg="#c0d5e1"
).pack(
    anchor="w",
    pady=(8, 0)
)

bob_public_value = tk.Label(
    bob_frame,
    text="",
    font=("Consolas", 16, "bold"),
    bg="#0f172a",
    fg="#c084fc",
    anchor="w",
    padx=10,
    pady=8
)

bob_public_value.pack(
    fill="x",
    pady=5
)

Ln: 1 Col: 0
```

```
CS_PRACS_GUI.py - C:\Users\info\OneDrive\Desktop\CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

    font=("Consolas", 16, "bold"),
    bg="#0f172a",
    fg="#22c55e",
    anchor="w",
    padx=10,
    pady=8
)

bob_secret_value.pack(
    fill="x",
    pady=5
)

# BUTTONS
button_frame = tk.Frame(
    root,
    bg="#0b1120"
)

button_frame.pack(
    pady=18
)

tk.Button(
    button_frame,
    text="@ EXCHANGE KEYS",
    command=calculate_exchange,
    font=("Arial", 11, "bold"),
    bg="#2563eb",
    fg="white",
    activebackground="#1d4ed8",
    activeforeground="white",
    relief="flat",
    padx=30,
    pady=10,
    cursor="hand2"
).grid(
    row=0,
    column=0,
    padx=8
)

Ln: 1 Col: 0
```

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```
CS_PRACS_GUI.py - C:\Users\info\OneDrive\Desktop\CS_PRACS_GUI.py (3.14.6)
File Edit Format Run Options Window Help

)

result_title = tk.Label(
    result_frame,
    text="READY",
    font=("Arial", 16, "bold"),
    bg="#111827",
    fg="#60a5fa"
)

result_title.pack(
    pady=(5, 2)
)

result_message = tk.Label(
    result_frame,
    text="Enter the parameters and start the key exchange.",
    font=("Arial", 10),
    bg="#111827",
    fg="#94a3b8"
)

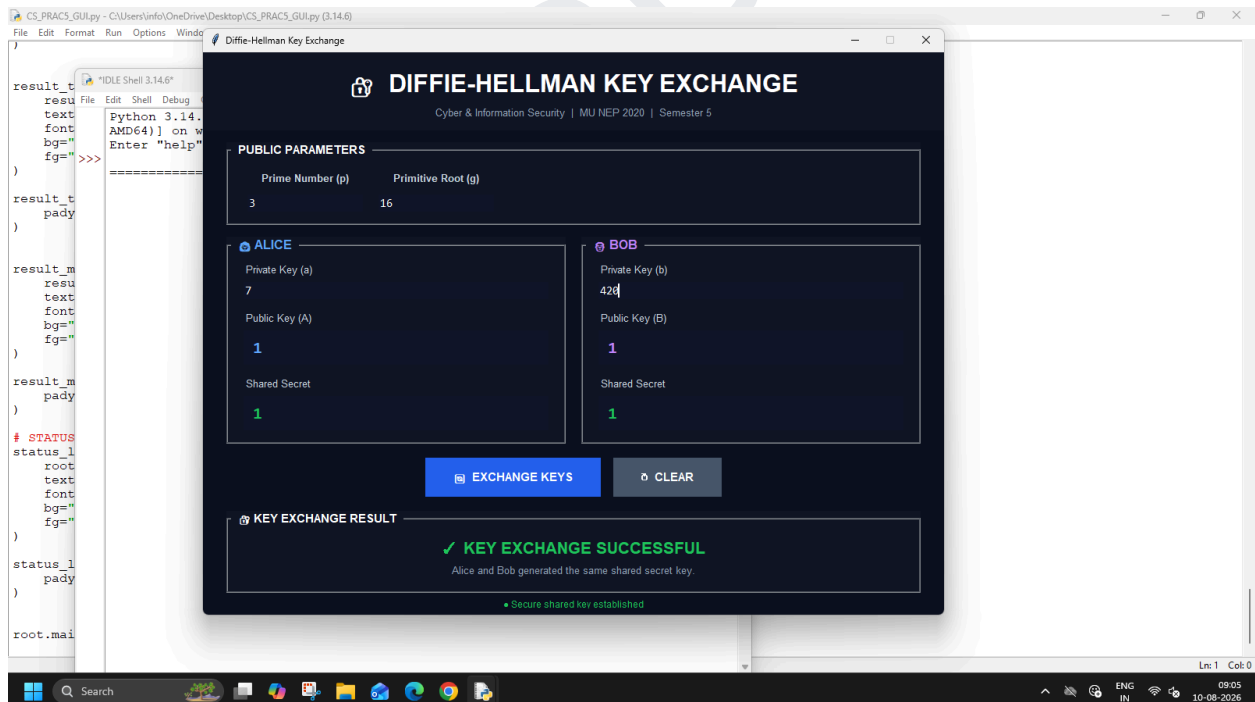
result_message.pack(
    pady=(0, 8)
)

# STATUS
status_label = tk.Label(
    root,
    text="● Ready",
    font=("Arial", 9),
    bg="#0b1120",
    fg="#94a3b8"
)

status_label.pack(
    pady=8
)

root.mainloop()
```

OUTPUT:



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